

SDG 6.4 Guided Challenge

Sustainable Water Allocation in the Alto Atoyac Basin Under Drought Scenarios

June 24, 2026

1 Guided Challenge Description

1.1 SDG Sub-Target

SDG 6 – Clean Water and Sanitation

Target 6.4: Increase water-use efficiency across all sectors and ensure sustainable withdrawals and supply of freshwater to address water scarcity.

1.2 Problem Statement

The Alto Atoyac Basin faces increasing pressure on water resources due to population growth, agricultural demand, industrial activity, groundwater extraction, and climate variability.

Under drought conditions, decision-makers must allocate limited water resources among competing users while protecting water-source sustainability and prioritizing human consumption.

1.3 Benchmark Quantity of Interest

The challenge minimizes the **Normalized Weighted Water Deficit (NWWD)**:

$$NWWD = w_U \sum_{j \in J} \frac{u_j^{urban}}{D_j^{urban}} + w_A \sum_{j \in J} \frac{u_j^{agri}}{D_j^{agri}}$$

where:

- u_j^{urban} is the unmet urban demand in municipality j ,
- u_j^{agri} is the unmet agricultural demand in municipality j ,
- D_j^{urban} is the total urban demand in municipality j ,
- D_j^{agri} is the total agricultural demand in municipality j ,
- w_U is the benchmark priority weight for urban/human consumption,
- w_A is the benchmark priority weight for agricultural demand.

For the benchmark instance:

$$w_U = 10, \quad w_A = 3$$

The deficit is normalized by total demand so that shortages are evaluated relative to the size of each demand category. This allows small and large municipalities, as well as urban and agricultural demand, to be compared more fairly.

These weights are provided by the organizers to ensure comparability across submissions. They explicitly encode priority: unmet urban/human-consumption demand is penalized more heavily than unmet agricultural demand, while agricultural impacts remain part of the objective.

1.4 Computational Bottleneck

The computational bottleneck is the constrained allocation of scarce water resources under:

- source-capacity constraints,
- drought-dependent availability,
- urban and agricultural demand,
- sector-priority weights,
- source-demand network costs.

As the number of sources, municipalities, demand categories, and allocation blocks increases, the search space grows combinatorially.

1.5 Computational Structure Assessment

The problem exhibits features commonly associated with quantum optimization:

- binary allocation decisions after discretization,
- combinatorial search space,
- graph/network structure,
- constrained optimization that can be encoded using penalty methods,
- transformation to a QUBO formulation.

A QUBO problem has the general form:

$$\min x^T Q x$$

where x is a binary decision vector and Q contains the objective coefficients and penalty terms.

1.6 Classical Baseline

The recommended baseline is a Mixed Integer Linear Programming (MILP) water-allocation model inspired by:

- Veintimilla-Reyes et al. [Veintimilla-Reyes et al., 2016],
- Fang et al. [Fang et al., 2011],
- Hsu and Cheng [Hsu and Cheng, 2002].

The MILP and the quantum implementation must optimize the same benchmark quantity of interest: NWWD.

1.7 MILP Baseline Formulation

Let:

- I be the set of water sources,
- J be the set of municipalities,
- x_{ij}^{urban} be the volume allocated from source i to urban demand in municipality j ,
- x_{ij}^{agri} be the volume allocated from source i to agricultural demand in municipality j ,
- A_i^{eff} be the drought-adjusted availability of source i ,
- D_j^{urban} be the urban demand of municipality j ,
- D_j^{agri} be the agricultural demand of municipality j ,
- u_j^{urban} and u_j^{agri} be unmet urban and agricultural demand,
- c_{ij} be the allocation cost from source i to municipality j .

The objective is:

$$\min \left[w_U \sum_{j \in J} \frac{u_j^{urban}}{D_j^{urban}} + w_A \sum_{j \in J} \frac{u_j^{agri}}{D_j^{agri}} + \lambda \sum_{i \in I} \sum_{j \in J} c_{ij} (x_{ij}^{urban} + x_{ij}^{agri}) \right]$$

where λ is an optional coefficient that controls the relevance of allocation cost relative to water deficit. For the required benchmark, participants may set $\lambda = 0$ or report the value used.

Subject to source capacity:

$$\sum_{j \in J} (x_{ij}^{urban} + x_{ij}^{agri}) \leq A_i^{eff} \quad \forall i \in I$$

urban demand balance:

$$\sum_{i \in I} x_{ij}^{urban} + u_j^{urban} = D_j^{urban} \quad \forall j \in J$$

agricultural demand balance:

$$\sum_{i \in I} x_{ij}^{agri} + u_j^{agri} = D_j^{agri} \quad \forall j \in J$$

and non-negativity:

$$x_{ij}^{urban} \geq 0, \quad x_{ij}^{agri} \geq 0, \quad u_j^{urban} \geq 0, \quad u_j^{agri} \geq 0$$

This formulation makes the restrictions explicit: water allocation cannot exceed drought-adjusted availability, and unmet urban and agricultural demand are tracked separately and normalized by their corresponding total demand.

1.8 Quantum Approach

The recommended quantum approach is:

Quantum Approximate Optimization Algorithm (QAOA)

Participants should discretize the MILP allocation variables into binary allocation decisions and formulate an equivalent QUBO representation of the benchmark problem.

1.9 QUBO Formulation Requirement

Participants should formulate a QUBO representation equivalent to the benchmark MILP model.

The QUBO formulation should preserve:

- the benchmark quantity of interest, NWWD,
- source-capacity constraints,
- urban and agricultural demand-balance constraints,
- allocation costs, if included.

Standard binary encoding and quadratic penalty techniques may be used.

Participants should document:

- their discretization strategy,
- binary encoding,
- penalty coefficients,
- resulting number of binary variables,
- feasibility of the solutions obtained.

Any equivalent QUBO formulation that preserves the original optimization objective and constraints is acceptable.

1.10 Evaluation

Participants should:

1. Use the benchmark instance provided in Annex A.
2. Implement the MILP baseline.
3. Reformulate the benchmark instance as a QUBO.
4. Implement QAOA or another quantum optimization method.
5. Compare against the classical baseline.

Metrics:

- NWWD,
- unmet urban demand,
- unmet agricultural demand,
- feasibility,
- runtime,
- scaling behaviour,
- number of binary variables.

1.11 Current Quantum Limitations

Current quantum hardware is not expected to outperform state-of-the-art MILP solvers for realistic instances of this problem.

The goal is not to demonstrate quantum advantage, but to assess whether the computational structure of drought-based water allocation is suitable for future quantum optimization methods.

1.12 Expected SDG Impact

Primary impact: SDG 6.4 – improved water allocation under scarcity.

Secondary impacts:

- SDG 11 – Sustainable Cities and Communities,
- SDG 13 – Climate Action.

Potential trade-offs:

- SDG 2 – Zero Hunger,
- SDG 8 – Decent Work and Economic Growth.

A Annex A – Benchmark Instance and Optional Data Expansion

A.1 Purpose

To reduce the time required for data preparation, the organizers provide a compact benchmark instance directly in this annex. Participants should use this benchmark instance for the required comparison between MILP and QAOA.

The benchmark values below are synthetic but inspired by publicly available demographic, agricultural, hydrological, and drought-related data sources. Their purpose is to provide a compact, reproducible optimization instance for the hackathon, not to represent official operational values for the Alto Atoyac Basin.

Participants may optionally expand the problem using public data sources listed at the end of this annex.

A.2 Benchmark Municipalities

Municipality	Urban Demand (hm ³ /year)	Agricultural Demand (hm ³ /year)
Puebla	95	8
San Andrés Cholula	14	5
Atlixco	9	18

These municipalities were selected to provide a representative mix of highly urbanized and agricultural demand profiles within the Puebla portion of the Alto Atoyac Basin.

All demand values are positive, which allows normalized deficits to be computed directly for both urban and agricultural categories.

A.3 Benchmark Water Sources

Water availability is treated at the **source level**. This means that aquifers or other water sources are modeled as supply nodes, while municipalities are modeled as demand nodes. This is a modeling abstraction: availability is associated with water sources, not directly with individual municipalities.

Source	Base Availability (hm ³ /year)	Source Type
Valle de Puebla Aquifer	80	Groundwater
Atlixco-Izúcar Aquifer	45	Groundwater

A.4 Benchmark Drought Scenarios

Scenario	Availability Factor
Normal	1.00
Moderate drought	0.80
Severe drought	0.60

Effective availability is computed as:

$$A_i^{eff} = \alpha A_i$$

where α is the drought availability factor.

A.5 Benchmark Source-Demand Network

Source	Municipality	Allocation Cost
Valle de Puebla Aquifer	Puebla	1.0
Valle de Puebla Aquifer	San Andrés Cholula	1.2
Valle de Puebla Aquifer	Atlixco	2.5
Atlixco-Izúcar Aquifer	Puebla	2.8
Atlixco-Izúcar Aquifer	San Andrés Cholula	2.0
Atlixco-Izúcar Aquifer	Atlixco	1.0

The allocation cost is a simplified benchmark value representing relative distance or delivery difficulty. It is not intended to represent actual infrastructure cost.

A.6 Benchmark Priority Weights

Demand Category	Weight
Urban / human consumption	10
Agricultural demand	3

A.7 Optional Crop-Level Expansion

If participants wish to expand the agricultural component, they may use the following benchmark crop-water requirements.

Crop Type	Water Requirement (m ³ /ha/year)
Maize	7000
Alfalfa	12000
Beans	4500
Wheat	5500
Barley	4500
Sorghum	6000
Oats	5000
Vegetables (generic)	8000
Potato	6500
Onion	7500
Chili Pepper	6500
Fruit Trees (generic)	9000

These values are benchmark assumptions for comparability and are not official crop-water requirements for the Alto Atoyac Basin.

A.8 Optional Data Sources for Scaling

Participants who wish to scale the benchmark instance may use the following public sources.

Municipal Boundaries and Locations

INEGI Marco Geoestadístico:

<https://www.inegi.org.mx/temas/mg/>

Population

INEGI Census 2020:

<https://www.inegi.org.mx/programas/ccpv/2020/>

Water Availability

CONAGUA Aquifer Availability:

https://sigagis.conagua.gob.mx/gas1/sections/Disponibilidad_Acuiferos.html

Agricultural Data

SIAP Agricultural Production:

https://nube.agricultura.gob.mx/cierre_agricola/

Water Rights, Optional

REPDA:

<https://app.conagua.gob.mx/consultarepda.aspx>

Drought Information

Mexican Drought Monitor, SMN:

<https://smn.conagua.gob.mx/es/climatologia/monitor-de-sequia/monitor-de-sequia-en-mexico>

B Annex B – Baseline and Benchmarking Strategy

B.1 Required Benchmark

All participants should report results for the benchmark instance in Annex A under at least:

- Normal scenario,
- Moderate drought scenario,
- Severe drought scenario.

B.2 Required Comparison

Participants should compare:

- MILP baseline,
- QUBO formulation,
- QAOA or another quantum/hybrid implementation.

B.3 Expected Outputs

Each submission should report:

- NWWD for each scenario,
- allocation decisions,
- unmet urban demand,
- unmet agricultural demand,
- normalized unmet urban demand,
- normalized unmet agricultural demand,
- feasibility,
- runtime,
- number of binary variables in the QUBO.

References

- H. Fang, Y. Cheng, and S. Yan. Optimization on water resource system operation policy during drought. *Journal of Water Resource and Protection*, 3:140–146, 2011. doi: 10.4236/jwarp.2011.32017.
- N.-S. Hsu and W.-M. Cheng. Network flow optimization model for basin-scale water supply planning. *Journal of Water Resources Planning and Management*, 128(2):102–112, 2002.
- J. Veintimilla-Reyes, F. Cisneros, P. Ochoa-Cueva, et al. Mixed integer linear programming (milp) approach to deal with spatio-temporal water allocation. *Procedia Engineering*, 162:221–229, 2016. doi: 10.1016/j.proeng.2016.11.045.